

Reducing Cardiovascular Disease in Indonesia

Health Impact, Costs, and Cost-Effectiveness of Clinical and Public-Health Interventions, 2025–2050

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Reducing CVD in Indonesia

Why this matters for Indonesia

Cardiovascular disease is Indonesia's largest health burden

Under current trends, a person reaching age 30 faces roughly a **14.5% chance** of dying from cardiovascular disease before age 70 — the CVD “40q30”. Most of this is **preventable or treatable**.

- Premature CVD deaths (ages 30–69) have risen for two decades and remain high.
- Two complementary levers:
 - **Clinical care** — prevent and better treat established disease.
 - **Population prevention** — tobacco, salt, trans-fat.
- This analysis weighs their **health impact and cost-effectiveness** together.

Source: Indonesia CVD model (calibrated to GBD 2023; UN World Population Prospects 2024). CVD 40q30 = probability of dying from cardiovascular disease between exact ages 30 and 70.

What this analysis set out to do

General aim

Estimate the **health impact and cost-effectiveness** of scaling up evidence-based **clinical** and **public-health** cardiovascular interventions in Indonesia through **2050**.

Aim 1 — Burden

Project the CVD burden and the premature-mortality indicator (CVD 40q30) under the **status quo** and under **alternative intervention scenarios**.

Aim 2 — Value for money

Estimate **deaths averted**, mortality reductions, **incremental costs**, and the **cost per death averted** for each intervention and package.

Aim 3 — Better together

Quantify the **added value of combining** clinical and public-health action, modeled as a single joint run rather than a sum.

Horizon 2025–2050; the “status quo” baseline is the no-new-intervention comparator against which every scenario is measured.

Executive summary — the headline numbers

Combining clinical + public health is the big prize

Together they could avert about **3.5M deaths** over 2025–2050, cut the CVD 40q30 in 2050 by **29%**, at about **US\$4,814 per death averted**.

- **All clinical interventions:** $\approx$ 2.0M deaths averted; CVD 40q30 down 20% by 2050.
- **All public-health policies:** $\approx$ 1.6M deaths averted at about US\$305 per death averted.
- **Most impactful single levers:** **Primary CVD prevention** and **Tobacco tax**.
- **Affordability:** the combined package costs about US\$4,814 per death averted.

All figures from the model's current-run cost/value outputs. Deaths averted are cumulative 2025–2050 across modeled cardiovascular and diabetes causes. Cost per death averted uses discounted incremental cost in market US\$ (price year 2023).

How the model works — in plain language

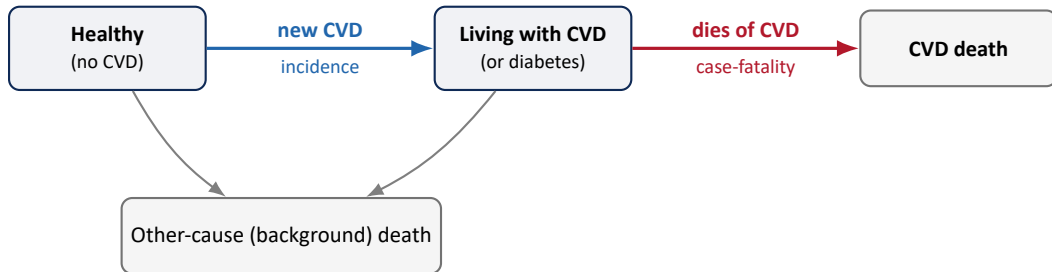
1. **People & demography.** Indonesia's population is projected by age and sex (UN World Population Prospects 2024).
2. **Starting health.** Disease and death rates are set to Indonesia's data (GBD 2023) and calibrated to match observed levels.
3. **Year-by-year simulation.** Each year people can develop cardiovascular disease or diabetes, survive, or die.
4. **Interventions.** Clinical and public-health scenarios change how many people develop disease and how many survive it.
5. **Outcomes.** The model reports deaths, healthy years (DALYs), premature-mortality (40q30), costs, and cost-effectiveness.

One consistent engine

Every scenario runs through the **same** projection, so results are directly comparable and the **combined** package is modeled as one run — not a sum of separate estimates.

Calibration fits the model to GBD estimates using a standard numerical optimiser. Demography and life expectancy use UN WPP 2024.

How interventions change the disease pathway



Fewer people fall ill

Primary prevention & *all* public-health policies lower **incidence**.

More people survive

Secondary prevention, treatment & tobacco policy lower **case-fatality**.

Background mortality

Other-cause deaths from demography; unchanged by these interventions.

No recovery/remission transition is modeled: people living with CVD remain at elevated risk (a conservative assumption).

Source: model engine, 06_run_scenarios_indonesia_fair.R.

The interventions

Clinical interventions (FAIR Choices)

Intervention	Target population	Model effect
Primary CVD prevention	Adults at cardiovascular risk	Incidence
Type-2 diabetes treatment	People with type-2 diabetes	Case-fatality
Heart-attack secondary prevention	Heart-attack (IHD) survivors	Case-fatality
Chronic heart-failure care	Chronic heart-failure patients	Case-fatality
Acute heart-failure care	Acute heart-failure patients	Case-fatality
Stroke secondary prevention	Stroke survivors	Case-fatality
RHD secondary prevention	Rheumatic heart disease patients	Case-fatality

Implementation scenario (all clinical interventions): coverage scales up **from a low base toward 80%** of the eligible population between 2025 and 2050. Effect sizes are drawn from the FAIR Choices intervention taxonomy; primary prevention lowers who develops CVD, while secondary prevention and treatment improve survival.

Source: workbook Selected_Interventions.

Public-health interventions (fiscal & regulatory policy)

Policy	Risk factor	Model effect	Implementation lever
Tobacco tax	Tobacco	Incidence + survival	Raise tax to 75% of retail price
Sodium policy package	Dietary salt/sodium	Incidence	Reformulation, labelling, media, procurement
Tobacco policy package	Tobacco	Incidence + survival	Clean-air law, mass media, ad ban

Policies reduce exposure (smoking, salt, trans-fat intake), lowering CVD **incidence**. **Tobacco** policies additionally improve **survival** after CVD. Packages are modeled as single joint runs of their component measures. Source: workbook *Selected_Interventions / Policy_Lever*s.

Data sources

Model component	Primary source	Years / version	Use in model
Population & demography	UN World Population Prospects	2024 revision	Population, background mortality, life expectancy
Cause-specific mortality & disease rates	GBD (Indonesia)	2023	Baseline burden & calibration targets
Disability weights (healthy-years lost)	GBD	2023	DALYs
Clinical intervention effects	FAIR Choices intervention taxonomy	current	Incidence & case-fatality effects
Public-health effect sizes	Published risk-factor evidence (in workbook)	current	Incidence & survival effects
Intervention & policy costs	Programme & policy unit costs (in workbook)	price year 2023	Budget impact & cost-effectiveness
International 40q30 comparison	WPP-scaled ASEAN comparison series	1995–2050	Benchmarking Indonesia vs. neighbour

Sources verified in the modeling pipeline and input workbooks. Specific effect-size and cost citations are documented in the input workbooks. Population: UN World Population Prospects 2024; GBD (GBD 2023 Cardiovascular Diseases and Risks Collaborators 2025; Schumacher et al. 2024).

Health impact and value for money

Clinical: impact and cost-effectiveness

Clinical intervention	Deaths averted (cumulative)	CVD 40q30 2050	40q30 red. 2050	Deaths red. 2050	Cost per death averted
Primary CVD prevention	1,322,000	7.6%	16.5%	10.9%	US\$919
Type-2 diabetes treatment	369,000	9.1%	0.3%	2.4%	US\$26,411
Heart-attack secondary prevention	141,000	8.9%	1.8%	0.8%	US\$21,376
Chronic heart-failure care	77,000	9.0%	0.9%	0.5%	US\$7,831
Acute heart-failure care	69,000	9.0%	0.6%	0.5%	US\$13,210
Stroke secondary prevention	31,000	9.1%	0.3%	0.2%	US\$56,216
RHD secondary prevention	6,000	9.1%	0.0%	0.0%	US\$24,755
All clinical (combined)	1,978,000	7.3%	19.8%	14.9%	US\$8,312

Cost per death averted is based on **discounted** incremental costs (3% per year) and assumes a linear scale-up from baseline coverage in 2025 to 80% coverage by 2050. Costs are reported in market US\$ (price year 2023) from the health system perspective.

Public health: impact and cost-effectiveness

Public-health policy	Deaths averted (cumulative)	CVD 40q30 2050	40q30 red. 2050	Deaths red. 2050	Cost per death averted
Tobacco tax	724,000	8.6%	5.2%	3.3%	US\$283
Sodium policy package	540,000	8.7%	4.0%	3.0%	US\$333
Tobacco policy package	435,000	8.8%	3.1%	2.0%	US\$272
All public health (combined)	1,649,000	8.0%	11.9%	8.0%	US\$305

Cumulative deaths averted during 2025–2050, relative to the status-quo baseline. The tobacco package combines smoke-free policies, health warnings and mass-media campaigns, advertising bans, and cessation support. The sodium package combines food reformulation, front-of-pack nutrition labelling, healthy public food-procurement policies, and behavior-change communication. Each package is modeled as a single joint scenario.

The combined package

Clinical + public health: better together

Package	Deaths averted (cumulative)	40q30 red. 2050	CVD 40q30 2050	Total cost (PV, discounted)	Cost per death averted
All clinical (combined)	1,978,000	19.8%	7.3%	US\$16.4B	US\$8,312
All public health (combined)	1,649,000	11.9%	8.0%	US\$503.2M	US\$305
Clinical + public health (combined)	3,453,000	29.2%	6.4%	US\$16.6B	US\$4,814

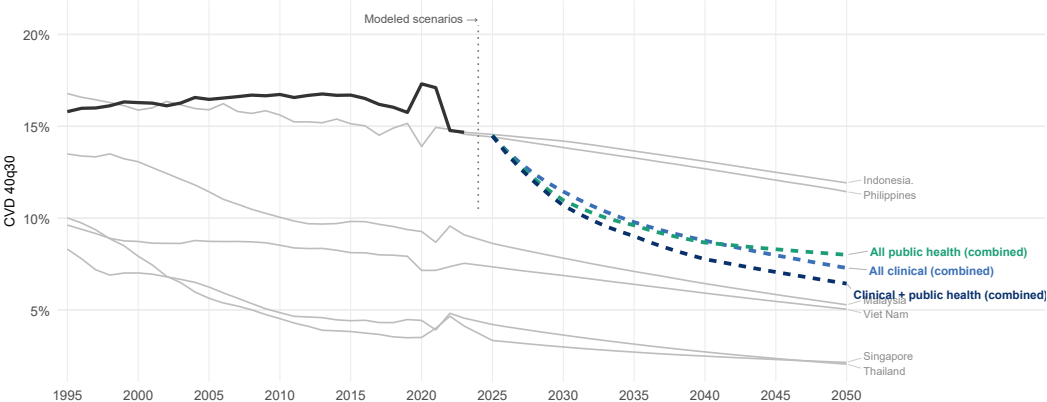
The combined package is modeled as one run — not a sum

Because clinical and public-health effects overlap, the joint scenario is projected in a single run. It averts about **3.5M deaths**, brings the 2050 CVD 40q30 down to **6.4%** (from 9.1% at baseline), at about **US\$4,814 per death averted**.

CVD premature mortality: trajectory and neighbours

CVD 40q30 in Indonesia: observed, projected, and neighbours

Probability of dying from CVD between ages 30 and 70



Indonesia observed history (solid, to 2023) and ASEAN comparators are WPP-scaled reference series.
Modeled intervention scenarios are dashed for 2025–2050.
ASEAN comparator trends use linear extrapolation for illustrative reference.

What it means for Indonesia

Policy implications

- **Biggest health gains come from doing both.** The combined package averts far more (~3.5M) than either family alone.
- **Public-health policies are exceptional value.** Population-wide fiscal/regulatory measures avert deaths at a fraction of clinical cost per death averted.
- **Clinical scale-up delivers large, durable gains** in survival and premature mortality.
- **Sequence for early wins.** Tobacco taxation and other prevention policies can start cutting risk quickly and cheaply, while clinical capacity scales up.
- **The 40q30 target is within reach.** The combined package bends Indonesia's premature-CVD-mortality curve down by about 29% by 2050.

These are model findings; translating them into policy requires judgement on feasibility, financing and delivery. Model results quantify the *potential*; they are not a delivery guarantee.

Limitations

- Results depend on **modeled epidemiological estimates** (GBD) and calibration; real burden may differ.
- **Effect sizes, future coverage, and unit costs are uncertain**; interventions are assumed to reach and sustain their targets.
- The disease model is **simplified** (healthy → living-with-CVD → death; no remission; no comorbidity interactions).
- Cost-effectiveness is expressed as **cost per death averted**; morbidity and productivity effects are not monetised here.
- Projections are **scenarios, not forecasts or guarantees**; they show what is achievable if interventions are delivered as modeled.

Three things to remember

1. Both, together

Clinical + public health averts **~3.5M deaths** by 2050 — more than either alone.

2. Prevention is affordable

Population policies avert deaths at about **US\$305** each.

3. A reachable target

Premature CVD mortality (40q30) falls **29%** by 2050 under the combined package.

Combined action against CVD could avert about **3.5M deaths** at about **US\$4,814 per death averted** through 2050.

Technical appendix

Appendix 1 — Scenario crosswalk

Family	Workbook scenario id	Executive label
Clinical (FAIR Choices)	all	All selected interventions (combined)
	I_CVD_PRIMARY	Primary prevention of CVD
	I_HF_ACUTE	Acute heart-failure management
	I_HF_CHRONIC	Chronic heart-failure management
	I_IHD_SECONDARY	Secondary prevention of ischemic heart disease
	I_RHD_SECONDARY	RHD secondary prevention
	I_STROKE_SECONDARY	Secondary prevention of stroke
	I_T2D_TREATMENT	Longitudinal treatment of type 2 diabetes
	baseline	Baseline (no new intervention)
Public health	all_public_health	All selected public-health interventions (combined)
	I_PH_SALT_POLICY	Salt policy package
	I_PH_TOBACCO_POLICY	Tobacco policy package
	I_PH_SALT_ENV	Supportive public-institution environment
	I_PH_SALT_FOPL	Sodium front-of-pack labelling
	I_PH_SALT_MEDIA	Sodium mass-media campaign
	I_PH_SALT_REFORM	Mandatory sodium reformulation
	I_PH_TOBACCO_TAX	Tobacco tax
	I_PH_TOB_AD_BAN	Tobacco advertising ban
	I_PH_TOB_CLEAN_AIR	Smoke-free / clean-air law
	I_PH_TOB_MEDIA	Tobacco mass-media campaign
Combined	all_clinical_public_health	All clinical + public-health interventions (combined)

Appendix 2 — Definitions, horizons and assumptions

Metrics

- **Deaths averted (cumulative)**
$$= \sum_{t=2025}^{2050} (\text{baseline deaths}_t - \text{scenario deaths}_t),$$

across modeled CVD & diabetes causes.
- **% reduction in deaths, 2050**
$$= 100 \times (\text{baseline} - \text{scenario}) / \text{baseline in 2050}.$$
- **CVD 40q30** = probability of dying of CVD between exact ages 30 and 70 (six CVD causes, both sexes, single-year life table).
- **% reduction in 40q30, 2050**: analogous to deaths.

Model. Discrete-time state-transition model; single-year ages 0–95; six CVD causes (IHD, ischemic stroke, intracerebral haemorrhage, hypertensive heart disease, rheumatic heart disease, cardiomyopathy & myocarditis) plus type-2 diabetes; calibrated to GBD 2023; demography and life expectancy from UN WPP 2024. **Data provenance:** every number is read from the current-run results contract (output/09_deck_results.rds) and the workbooks' literal CVD 40q30 series; scenario identity follows the Scenario_Catalog contract (binding include_flag). No Excel recalculation is used.

Costing & cost-effectiveness

- Horizon 2025–2050; cost discount rate 3%/yr; price year 2023.
- Costs in **market US\$**; cost perspective: health system.
- **Cost per death averted** = cumulative **discounted** incremental cost ÷ cumulative **undiscounted** deaths averted.
- A “Dominant” scenario averts deaths while lowering cost; “not defined” means no deaths averted.

Appendix 3 — References

GBD 2023 Cardiovascular Diseases and Risks Collaborators. 2025. “Global, Regional, and National Burden of Cardiovascular Diseases and Risk Factors in 204 Countries and Territories, 1990–2023.” *Journal of the American College of Cardiology* 86 (22): 2167–2243.

Schumacher, Austin E., Hmwe Hmwe Kyu, Catherine M. Antony, Aleksandr Y. Aravkin, and Gulrez. 2024. “Global Age-Sex-Specific Mortality, Life Expectancy, and Population Estimates in 204 Countries and Territories and 811 Subnational Locations, 1950–2021, and the Impact of the COVID-19 Pandemic: A Comprehensive Demographic Analysis for the Global Burden of Disease Study 2021.” *The Lancet* 403 (May): 1989–2056.
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